

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-142

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF ESCALON
ESCALON WASTEWATER TREATMENT PLANT
SAN JOAQUIN COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. The City of Escalon (hereafter Discharger) submitted a Report of Waste Discharge, dated 29 February 2000, and supplemental information dated 30 March 2000, for an expansion of the industrial portion of its wastewater treatment plant (WWTP).
2. The wastewater treatment facility is at 25100 West River Road, Escalon, in Sections 17 and 20, T2S, R9E, MDB&M, as shown in Attachment A, which is attached hereto and made part of this Order by reference. The facility is on Assessor's Parcel Numbers 247-090-36 and 247-090-38, which are both owned by the City of Escalon.
3. Waste Discharge Requirements Order No. 95-170, adopted by the Board on 23 June 1995, prescribes requirements for discharge of treated effluent from the City of Escalon WWTP to evaporation and percolation ponds on the property. Because of increased flows to the WWTP, and facility improvements at the site, Order No. 95-170 is not adequate and requires amendment.
4. The WWTP treats industrial wastewater generated by vegetable processing industries and domestic wastewater generated by the City of Escalon. The treatment process consists of screening and discharge to mechanically aerated treatment ponds, followed by discharge to evaporation/percolation ponds. The ponds, and proposed improvements, are shown in Attachment B, which is attached hereto and made part of this Order by reference.
5. The industrial and domestic wastewater flows are delivered to the plant in separate pipes, are treated in separate ponds, and are discharged to separate evaporation/percolation ponds. Stormwater from a limited portion of the City of Escalon is piped to the facility and is discharged to the industrial ponds.
6. The Discharger treats domestic wastewater in five treatment ponds, all of which are mechanically aerated. The domestic wastewater is disposed in two evaporation/percolation ponds (Nos. 21 and 24). Pond No. 21 provides 4.4 acres of surface area and 11.3 million gallons of storage capacity. Pond No. 24 provides 4.7 acres of surface area and 11.1 million gallons of storage capacity.
7. Prior to recent upgrades, the treatment plant was designed to treat domestic wastewater average dry weather flows of up to 0.55 million gallons per day (mgd) and peak wet weather flows of 1.0 mgd.

8. A report prepared by Dewante and Stowell titled, "*Wastewater Treatment Facilities Improvement and Expansion*," dated April 1990, identified improvements at the WWTP that could increase capacity and treatment. Improvements that were implemented included construction of Pond Nos. 21 and 24, addition of mechanical aeration, and piping improvements. The improvements resulted in additional storage capacity of 22.4 million gallons and 9.1 acres of surface area to the existing domestic wastewater treatment system. Based on the information in the Dewante and Stowell report, the domestic wastewater treatment facility is designed to treat 0.90 million gallons a day.
9. The Discharger does not disinfect effluent prior to discharge to the evaporation/percolation ponds.
10. The Discharger treats industrial wastewater in four treatment ponds, all of which are mechanically aerated. The industrial wastewater is then discharged to ten evaporation/percolation ponds (Nos. 10 through 19). The approximate surface area and storage capacity is presented below:

<u>Pond Number</u>	<u>Surface Area (acres)</u>	<u>Storage Capacity (millions of gallons)</u>
1	2.8	5.4
2	1.7	1.7
3	1.8	1.8
4	2.3	3.0
10	1.1	2.2
11	2.1	6.1
12	1.8	4.7
13	1.8	2.8
14	1.3	2.0
15	1.0	1.5
16	2.1	2.7
17	2.9	3.7
18	2.7	3.3
19	2.7	3.3
Totals	28.1	44.2

Data from City of Escalon, *Treatment Plant Storage Map*, dated 1/17/97.

11. The treatment plant was designed to treat industrial wastewater average dry weather flows up to 2.0 mgd and peak wet weather flows of 3.0 mgd. The monthly average dry weather maximum industrial flow rate was exceeded in August, 1999; the maximum daily industrial flow rate was exceeded in September and October, 1999.
12. The Discharger allows the industrial wastewater ponds to dry completely prior to initiation of the industrial wastewater generation season. The ponds are disced and ripped annually to maintain percolation rates.

13. Industrial dischargers consist of Escalon Premier Brand (tomato canner) and Eckert Cold Storage (frozen pepper processor). The industrial dischargers' processing season is from approximately May through December. The industrial dischargers presently screen their wastewater to remove solids prior to discharge to the WWTP.
14. Industrial wastewater is characterized by high concentrations of biochemical oxygen demand. The Discharger has not been previously required to analyze for total dissolved solids (TDS) concentrations. Escalon Premier Brands added tomato peeling equipment in 1999 which resulted in additional flow and increased biochemical oxygen demand.
15. Based on the Discharger's self-monitoring data for 1999, the flow rate and biochemical oxygen demand of influent at the WWTP headworks is presented below:

<u>Month</u>	<u>Domestic Wastewater System</u>		<u>Industrial Wastewater System</u>	
	<u>Flow (mg)</u>	<u>BOD¹ (mg/l)</u>	<u>Flow (mg)</u>	<u>BOD¹ (mg/l)</u>
January 1999	21.2	282	0.75	NA
February 1999	19.2	106	0.21	NA
March 1999	20.1	215	0.25	NA
April 1999	19.5	114	0.15	NA
May 1999	19.4	102	0.60	NA
June 1999	20.4	147	7.54	712
July 1999	20.4	181	42.6	505
August 1999	21.3	105	79.7	610
September 1999	18.6	110	93.3	1,380
October 1999	21.9	131	98.7	750
November 1999	21.2	71	32.4	316
December 1999	21.3	75	5.2	NA

¹ 5-day Biochemical Oxygen Demand.

NA denotes Not Analyzed.

Data from City of Escalon, "1999 Annual Wastewater Treatment Report," February 2000.

16. Both industrial dischargers are required to sample their wastewater individually. Flow rates and biochemical oxygen demand data for the 1999 processing season are presented below:

<u>Constituent</u>	<u>Units</u>	<u>Escalon Premier Brand</u>	<u>Eckert Cold Storage</u>
BOD ₅	mg/l	<u>Average/peak</u> 1,151/1,900	<u>Average/Peak</u> 1,214/3,000
Flow Rates			
July 1999	mgd	1.8/2.0	0.5/1.0
August 1999	mgd	2.2/2.3	0.4/0.5
September 1999	mgd	2.5/3.0	0.7/0.9
October 1999	mgd	2.3/2.7	0.7/0.9
November 1999	mgd	2.1/2.3	0.7/0.8

Data from Nolte Report, "Wastewater Treatment Facilities Improvements," March 2000.

17. A hydraulic analysis of the industrial wastewater portion of the WWTP presented in the RWD indicates that industrial wastewater flows have reached the WWTP's hydraulic capacity. Reclamation is not presently performed; however, the Discharger is investigating reclamation alternatives.
18. The Discharger is in the process of improving the industrial WWTP. The improvements will provide more capacity and better treatment. Improvements include deepening the existing treatment ponds, three new evaporation/percolation ponds (Nos. 20, 22, and 24), piping and pond configuration improvements, addition of mechanical aerators, and reconfiguration of existing aerators. Pond No. 22 will provide 5.6 acres of surface area and 5.1 million gallons of storage capacity. Pond No. 23 will provide 5.0 acres of surface area and 4.7 million gallons of storage capacity. Pond No. 20 will be constructed as an enlargement of Pond No. 12; Pond No. 12/20 will provide 4.0 acres of surface area and 11.7 million gallons of storage capacity. The construction activities are scheduled to be completed by July 2000, in time for the beginning of the industrial discharger's processing season. A summary of the pond configuration improvements is presented below:

	Pond Area	Pond Volume	Hydraulic Capacity	
<u>Condition/Improvement</u>	<u>Acres</u>	<u>MG</u>	<u>MGD</u>	<u>MG/year</u>
Existing Conditions	19.5	32.3	2.0	221
<u>Recommended Improvements</u>				
Deepen Treatment Ponds	NA	3.1	NR	NR
Expand Pond 12 to Incorporate Pond 20	2.2	7.0	0.27	29
Remove Berm between Ponds 13 and 16	0.3	0.5	0.04	3
Remove Berm between Ponds 14 and 17	0.3	0.5	0.04	3
Remove Berm between Ponds 15 and 18	0.3	0.5	0.04	3
Construct Pond 22	5.6	5.1	0.54	58
Construct Pond 23	5.0	4.7	0.48	52
Subtotal of Improvements	13.7	18.3	1.41	148
Total (Ext. + Rec. Improvements)	33.2	50.6	3.41	370

Data from Nolte Report, "Wastewater Treatment Facilities Improvements," March 2000.

MG denotes million gallons. MGD denotes million gallons per day.

NA denotes not applicable. NR denotes not reported.

Ext. + Rec. denotes the sum of existing and recommended improvements.

19. Escalon Premier Brands is in the process of retaining a wastewater engineering firm to identify means to reduce the BOD concentrations of its wastewater and to investigate opportunities for recycling of wastewater at the plant.
20. Three groundwater monitoring wells exist at the facility. The wells are screened in the first saturated interval and are currently sampled annually. Well W-1 is upgradient of the

wastewater treatment ponds; Wells W-2 and W-3 are adjacent to the future Ponds 22 and 23. None of the wells are directly downgradient of the treatment or percolation/evaporation ponds. Based on groundwater samples which were collected from April 1997 to January 2000, average characterization of the water quality is:

<u>Well</u>	<u>Electrical Conductivity</u>	<u>Nitrate (as nitrate)</u>
W-1	780 mg/l	5.4 mg/l ¹
W-2	566.7 mg/l	20.8 mg/l
W-3	785 mg/l	43.1 mg/l

Data from Discharger's self-monitoring reports.

¹ Analytical detection limit used to calculate average value.

21. Based on the groundwater monitoring performed at the site, groundwater exists approximately 37 feet below ground surface. Groundwater flows to the west-southwest.
22. In order to determine compliance with the groundwater and surface water limitations contained herein, these WDRs contain a time schedule for preparation of a groundwater evaluation report.
23. Objectionable odor complaints related to the facility were documented in October 1996, November 1997, and September 1999. All the complaints were related to the industrial wastewater system.
24. The WWTP is on the north side of the Stanislaus River. Surrounding land uses are primarily agricultural. A golf country club and private residences exist south of the Stanislaus River.
25. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which designates beneficial uses and water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
26. The site lies within the San Joaquin Valley Floor Hydrologic Unit No. 535.10, as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
27. Surface water drainage is to the Stanislaus River. The beneficial uses of the Stanislaus River are municipal, agricultural water supply for irrigation and stock watering; contact recreation, canoeing, and non-contact recreation; warm and cold freshwater habitat; cold water migration; warm and cold water spawning; and wildlife habitat. The potential beneficial uses of the Stanislaus River are municipal and domestic water supply.
28. The beneficial uses of the underlying groundwater are municipal, domestic, and industrial supply.

29. The Board has considered anti-degradation pursuant to State Board Resolution No 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides a timeline for data collection to determine whether the discharge will cause an increase in groundwater constituents above that of background levels. If the discharge is causing such an increase, then the Discharger may be required to cease the discharge, line the ponds, implement source control, change the method of disposal, or take other action to prevent groundwater degradation.
30. The Discharger prepared a Negative Declaration in accordance with the California Environmental Quality Act (Public Resource Code Section 21000 et seq.) and the State CEQA Guidelines. The CEQA document was approved on 9 May 2000 by the City of Escalon City Council.
31. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
- a. The Board is issuing waste discharge requirements,
 - b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a designated or hazardous waste.
32. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."
33. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
34. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 95-170 is rescinded and the City of Escalon, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
4. The discharge shall not cause the degradation of any water supply.
5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15), or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications:

1. For the domestic WWTP, the monthly average dry weather flow shall not exceed 0.90 mgd. The maximum daily flow shall not exceed 1.0 mgd.
2. For the industrial WWTP, the monthly average dry weather flow shall not exceed 2.0 mgd. The maximum daily flow shall not exceed 3.0 mgd. These industrial flow limits will remain in effect until the industrial wastewater treatment facilities improvements are completed, and a technical report (as described in Provision F.1) has been submitted and approved by the Executive Officer. Upon approval of the technical report, the Discharger's flow limits will be allowed as presented in Discharge Specification No. B.3.
3. Upon approval of the technical report required by Provision F.1, flow rates at the industrial WWTP may increase as follows: the monthly average dry weather industrial discharge flow to the industrial WWTP shall not exceed a maximum average daily flow over any 30-day period of 3.4 mgd, and the total industrial discharge over the processing season shall not exceed 370 million gallons.
4. The effluent discharged to the domestic evaporation/percolation ponds shall not exceed the following limits.

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅	mg/l	40 ¹	80
Settleable Matter	ml/l	0.2 ¹	--

¹ If the reported concentration is greater than the criterion presented, a confirmation sample shall be collected for analysis within seven days of receiving the analytical data.

5. The effluent contained in the industrial evaporation/percolation ponds shall not exceed the following limits.

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>
BOD ₅	mg/l	150

6. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment facility.
7. As a means of discerning compliance with Discharge Specification No. B.6, the dissolved oxygen content shall not be less than 1.0 mg/l in any pond at any time, as measured at a point as far as practical from the pond inlet and within one foot of the water surface.
8. The discharge to conveyance systems, settling basins, ponds, or land disposal areas not adequately maintained to prevent off-site odor nuisance, fly breeding, or mosquito breeding is prohibited.
9. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
10. The pond system shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, seasonal ancillary inflow, and infiltration during the wet season. Design seasonal precipitation shall be based on total annual precipitation using a return of 100 years, distributed monthly in accordance with historical rainfall patterns.
11. Pond freeboard shall never be less than two feet in any pond as measured vertically from the water surface to the upper surface of the lowest adjacent dike or levee.
12. On or about **1 November** each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specifications No. B.10 and B.11.

C. Solids Disposal Requirements:

1. Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.

3. Use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.
4. If the State Water Resources Control Board and the Regional Water Resources Control Board are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger shall comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Groundwater Limitations:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed 2.2/100 ml over any 7-day period.

E. Surface Water Limitations:

The Discharger shall not cause the Stanislaus River downstream of the evaporation/percolation ponds to contain waste constituents in concentrations statistically greater than background (upstream) surface water quality. Background surface water quality shall be determined when the required monitoring program provides sufficient data.

F. Provisions:

1. As described in Discharge Specification B.2, the Discharger shall submit a technical report which describes the implemented WWTP improvements. The report is required at least **15 days prior** to the Discharger's desire to increase the flow rates as described in Discharge Specification B.3.
2. By **1 January 2001**, the Discharger shall submit a technical report, prepared under the supervision of a registered engineer, which identifies opportunities to reduce, reuse, and recycle wastewater. The report shall identify sources of TDS in industrial wastewater, and recommend ways to reduce TDS. The report shall also evaluate reclamation alternatives both at the industrial discharger facilities and the WWTP. The report shall present a tabulation of all data used in evaluating the industrial wastewater flow.
3. By **1 January 2001** Discharger shall submit a Sludge Management Plan which describes the annual volume of sludge generated by the facility and specifies the proposed testing and disposal practices.
4. By **31 May 2001**, the Discharger shall submit a report, pursuant to Section 13267 of the California Water Code, indicating whether the discharge to the ponds has caused, or is

likely to cause, constituent concentrations in groundwater to exceed background concentrations. The report shall be prepared under the supervision of a Registered Geologist or Registered Engineer. If monitoring data indicates wastewater has degraded groundwater quality beyond background concentrations, the Discharger shall submit a Wastewater Disposal Mitigation Plan (WDMP) within 90 days of request by the Executive Officer. The WDMP shall address the magnitude and extent of groundwater contaminants, evaluate contaminant control alternatives, evaluate appropriate effluent limits, and select a preferred contaminant control alternative. The selected contaminant control alternative must comply with State Water Resources Control Board Resolution No. 68-16, Title 27 CCR, and the most recent Basin Plan. The WDMP shall include a proposed project schedule for design and construction.

5. The Discharger shall comply with Monitoring and Reporting Program No. 5-00-142, which is part of this Order, and any revisions thereto, as ordered by the Executive Officer.
6. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which is attached hereto and made part of this Order by reference. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)".
7. If the Discharger wishes to increase daily or monthly flow limits at the industrial WWTP above what is allowed in this Order, then it must submit an RWD. A complete RWD shall be submitted at least 120 days prior to the anticipated increase in discharge.
8. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.
9. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
10. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.
11. The Discharger shall submit to the Board on or before each compliance report due date, the specified document or, if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is being reported, then the Discharger shall state the reasons for such noncompliance and provide an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board in writing when it returns to compliance with the time schedule.
12. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders

requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

13. A copy of this Order shall be kept at the discharge facility for operating personnel. Key operating personnel shall be familiar with its contents.
14. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 16 June 2000.

GARY M. CARLTON, Executive Officer

Attachments
TRO: 6/16/00